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A Retinoid Antagonist Attenuates Bone Growth Inhibition Caused by Growth Plate Injury in Mice

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ABSTRACT

Growth plate injury, which may trigger growth plate fusion or arrest, is a common occurrence in pediatric orthopaedics that can lead to angulation of the limb and limb length discrepancy. Growth plate injuries are currently treated palliatively, with surgical intervention when severe limb length discrepancy or angulation is found during follow-up visits. The cellular changes in the chondrocytes in injured growth plate have not been fully elucidated, and understanding these factors will provide important insights for the development of therapies that promote growth plate repair and prevent growth arrest. In this study, we refined a mouse growth plate injury model and demonstrated that needle-induced growth plate injury resulted in different healing outcomes depending on the size of the needle used. The 30G needle-induced injury was repaired without significant sequelae, while 25G injury induced bone bridge and limb length discrepancy. In the non-healing model, the growth plate showed severely reduced bone-forming activity rapidly after the injury, with a decreased number of osteoclasts in the chondro-osseous junction in the metaphysis. We found that the injured growth plate stimulated retinoid signaling evidenced by the upregulation of retinoid target molecules, and that treatment with 7C, a retinoic acid receptor gamma antagonist, ameliorated decreases in bone forming activity and osteoclast formation, and partially rescued injury-induced growth inhibition. The results of this study suggest that changes in the growth plate immediately post-injury should be considered as a therapeutic target for growth plate repair and that retinoid antagonists may be viable pharmacological interventions for this purpose.

1 | Introduction

Fractures are common orthopaedic problems in children, with approximately 15-30% involving growth plate (GP, also called physis) [1, 2]. The GP is the hub of endochondral ossification and is responsible for the growth of the long bones [3, 4]; however, an injury to the GP may impair its function [1, 5]. The failure of functional recovery in the injured GP may trigger premature arrest or fusion, resulting in angulation of the limb and/or a limb length discrepancy (LLD) [5, 6]. These conditions

can subsequently impair smooth limb movement associated with pain, and further increase the risk of osteoarthritis [6].

The location of fracture within the GP is a key factor in the prognosis of growth disturbances [1, 5]. Fractures across a portion or full length of GP with propagation through the epiphyseal ossification center and into the metaphysis result in a higher risk of growth arrest [5]. In such cases, fracture communication between the epiphysis and metaphysis may allow for the formation of an osseous bridge (bone bridge or

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bony bar), which may mechanically disturb the longitudinal bone growth [1].

The treatment of GP injuries is initially palliative noninvasive procedures, and then during follow-up of GP injuries, surgical interventions may be required to reduce LLD and restore anatomical alignment of the joints, epiphysis, and growth plates [6]. Correction of limb angulation or LLD can be a significant burden for patients. To date, most clinical and preclinical studies have primarily focused the bone bridge as a therapeutic target [7–9]; however while the surgical removal of the bone bridge may ameliorate growth arrest, the outcome may depend on how well the GP retains its function. Understanding the cellular changes in chondrocytes in early stages of GP injury is important for stimulating GP repair and preventing growth arrest.

Chondrocytes in the GP undergo a series of functional changes, organizing distinct layers named resting, proliferating, prehypertrophic and hypertrophic zones [3, 4]. GP chondrocytes undergo active proliferation, the abundant production of cartilage matrix, and terminal differentiation characterized by hypertrophy, matrix degradation and calcification. Terminally differentiated chondrocytes and their calcified matrix are eventually invaded by blood vessels and osteoclasts, and subsequently replaced with bone. The rate and extent of phenotype changes in GP chondrocytes are sophisticatedly controlled by various factors including vitamins, hormones, growth factors and cytokines [10, 11].

When the GP is injured, chondrocytes respond to the injury and attempt to repair the defect. In prior studies, the GP was shortened 10–12 weeks post-injury in the proximal tibia of rabbits [12] and 7 weeks post-injury in the distal femur of rats [13] as determined by radiographical or histological evaluation. In contrast, Drenkard et al. [14] reported that the GP was locally thickened at the site of the injury in the distal femoral GP as early as 1 week post-injury, and the thickening was correlated with the disturbance of bone growth. Histological evaluation in the aforementioned study revealed thickening of the hypertrophic and resting zones, with a simultaneous decrease in the ratio of the proliferating zone [14]. Recent studies have demonstrated that skeletal stem cells in the resting zone propagate 3 days post-injury and contribute to the repair process in a repairable growth plate injury model in mice [15]. However, it remains unclear which changes in molecular pathways cause the disorganization and dysfunction of injured GPs, leading to growth arrest. To answer this question, we need to determine the spatiotemporal changes of injured GPs at the cellular and molecular levels using a rigorous and reproducible GP injury model. In this study, we refined a murine GP injury model that minimizes damage to surrounding tissues to reduce the complexity of the surgical procedure and to mitigate the indirect effects of other tissue reactions, and investigated early changes in the injured GP.

Controlling retinoid signaling is important for the regulation of cartilage development and GP function [16]. RAR γ dominantly mediates retinoid signaling in the GP [17]. Retinoid signaling activity is low in most regions of the GP while it is elevated in the terminally differentiated chondrocytes in the hypertrophic

zone [18]. Excessive activation of retinoid signaling in the GP impairs proliferation [19] and accelerates hypertrophy of chondrocytes [20]. Of note, an association was found between the upregulation of retinoid signaling and injury in a variety of tissues, including the skeletal muscle, heart, liver, and lungs [21, 22]. We, here, examined whether retinoid signaling was also upregulated in the injured GP. As we previously demonstrated that a retinoid antagonist had an inhibitory action on growth disturbance caused by a cancer drug [23], we aimed to determine whether retinoid antagonists also have an inhibitory effect on GP injury-induced growth inhibition.

2 | Methods

2.1 | Animals

All animal experiments were conducted according to the protocol approved by the Institutional Animal Care and Use Committee of the Children's Hospital of Philadelphia and University of Maryland, Baltimore. A total of 201 mice were used: 39 male and 47 female C57BL/6J mice for characterization of GP injury, and 45 male and 70 female C57BL/6J mice for 7C treatment experiments. GP injuries were induced in the right proximal tibial GP in 3.5 to 4-week-old C57BL/6J mice (Jackson Laboratory, Bar Harbor, ME, USA). Mice received intraperitoneal injections of 5-ethynyl-20-deoxyuridine (EdU, 50 mg/kg, Sigma-Aldrich Inc. St. Louis, MO, USA) 2 h before euthanasia to monitor cell proliferation and/or alizarin complexon (30 g/kg, Sigma-Aldrich Inc.) 2 days before euthanasia. The tibias were harvested 1 day to 5 weeks after surgery, with the mice randomly divided into control and drug groups. Drug response was assessed based on the differences in bone length (primary outcome), bone formation activity, bone bridge formation (secondary outcome), and cellular changes. Animal treatment and sample analyses were performed in an unblinded manner.

2.2 | Preparation and Systemic Administration of Retinoids

Retinoic acid receptor gamma (RAR γ) antagonist 7C was synthesized by Atomax Chemicals (Shenzhen, China) according to the world patent application (WO 2005/066115 A2). Stock solutions of 7C in dimethyl sulfoxide (DMSO, D2650; Sigma-Aldrich Inc.) were stored at -30°C under argon. Before administration, aliquots of the stock solution were mixed with a three-fold volume of corn oil (C8267; Sigma-Aldrich Inc.) and orally administered to the mice (1 mg/kg). Vehicle control mice received 10 μL DMSO plus 30 μL corn oil in the same manner. Treatment was started 1 day after the induction of the GP injury and continued daily for 7 days.

2.3 | Statistical Analysis and Sample Size

Sample size was calculated using Power & Sample Size Calculator (null-hypothesis statistical test) provided by GIGA calculator (<https://www.gigacalculator.com/>) with power 0.80,

$\alpha = 0.05$, the expected difference ranging from 20% to 40%, expected standard deviation from 20% to 30%, and was revised according to the data. Sample size was $n = 6-7$ for bone length analysis, $n = 3-7$ for histology and histological staining (EdU incorporation, AR labeling, osteoclast staining, in situ hybridization, and immunofluorescence staining), and $n = 6-7$ for drug treatment experiments. The results were analyzed using Prism 10 version 10.0.1 (GraphPad Software, Boston, MA, USA). The Mann-Whitney U test or Wilcoxon matched-pairs test was used to compare two groups while the Kruskal-Wallis test was used to compare multiple groups. The threshold for significance for all tests was set at $p < 0.05$.

Additional methods are described in Supplemental Information.

3 | Results

3.1 | The GP Injury Model

Animal models for GP injuries usually involve creating a drill-hole defect throughout the full depth of the GP in the tibia or femur of rabbits [12], rats [8, 13], or mice [14]. This injury may be made from the diaphysis or metaphysis to the end of the epiphyseal bone [8, 12, 13, 24] or the articular cartilage to the metaphysis [14]. These models create a clinical mimic of the GP and are classified as Salter-Harris type IV [5], leading to bone bridge formation and bone growth disturbances. However, this procedure requires special skills to control the location, length, and angle of the drill-holes. Surgery may produce inconsistencies such as varying degrees of size and location of damage to the articular cartilage, GP, and bone, making it difficult to elucidate the detailed cellular and molecular changes which occur during GP injury, repair, and arrest. To minimize damage to the bone, articular cartilage and joint tissues, and to create a reproducible defect on the GP, we created a transverse defect by penetration of a needle into the GP of the proximal tibia from the medial to lateral site (Figure 1A,B and Figure S-1). Figures 1C and 1D show images of the proximal tibial GP before and after needle insertion. Changing the size of the needle allowed us to control the size of the defect. The penetration of a 25-gauge (25 G) needle made a full thickness defect in the proximal tibial GP (Figure 1E), and induced bone bridge formation (Figure 1F, histology; Figure 1G,H, μ CT, red).

The effects of the different needle sizes were also investigated, showing that the injuries made with the 18 G and 25 G needles inhibited the growth of the affected bone (Figure 1I, 18 G and 25 G) and induced consistent and significant LLDs (Figure 1J, 18 G and 25 G) whereas the injuries made with the 30 G needles did not (Figure 1I,J, 30 G). The outer diameters of the 18 G, 25 G, and 30 G needles are approximately 1.2, 0.5, and 0.3 mm, respectively. The width of the GP of 3-4 week old mice is approximately 0.3-0.4 mm [25]. Therefore, the 30 G needle was not large enough to create a full-thickness defect in the GP that induced growth disturbance, meaning the affected GP was likely repaired. Injuries with a 25 G needle resulted in LLDs in both females and males (Figure 1K), and sex differences in the extent of bone shortening were not detected (Figure 1L). Figure 2 shows the histology of the GP injured with a 25 G needle. Fibrous tissue appeared at the injury site 2 days post-injury,

communicating between the epiphyseal and metaphyseal bones (Figure 2A). At 1 week post-injury, localized widening of the damaged growth plate was observed (Figure 2C). The osteoid formed between the GP by 1 week post-injury (Figure 2C) and expanded 3 weeks post-injury creating bone bridge (Figure 2E), which was connected to the trabecular bone in the metaphysis 5 weeks post-injury (Figure 2G). The GP of the contralateral limb after sham surgery showed normal histology (Figure 2B,D,F and H). In contrast to the 25 G injury group, the GP injured with a 30 G needle did not form a bone bridge 3 weeks post-injury: Five of six samples showed normal histological findings (Figure S-2A), and the remaining one showed evidence of injury but no bone bridge (Figure S-2B).

3.2 | Early Changes in the Injured GP

Animal studies on GP injury have largely focused on bone bridge formation, which occurs within a certain period of time post-injury [7, 8]. Little information is available on how GPs rapidly change their functions. Using our growth injury model, we investigated the early changes in the injured GP. Proliferation was assessed by analyzing EdU uptake 1 and 3 weeks post-injury. No significant difference in EdU uptake was detected 1 week post-injury (Figure 3A,B, and E, 1 week), whereas 3 weeks post-injury, EdU uptake decreased in the injured GP (Figure 3C,D, and E, 3 weeks). The GP chondrocytes except for chondroprogenitors in the resting zone, are rapidly transitioned from a proliferating phase to a hypertrophic phase and disappear from the GP. To examine differences in cell turnover, proliferating GP chondrocytes were labeled with EdU before surgery, and the distribution of the EdU-labeled cells was examined 4 days post-injury. The number of EdU-labeled cells in the hypertrophic zone was significantly higher in the injured than that in the contralateral GP (Figure 3F-J), suggesting that cell turnover was slower in the injured GP.

This finding suggests that the GP function is acutely impaired by injury. To examine bone forming activity in the primary spongiosa, calcification was visualized by alizarin complexon (AR) labeling. Mice underwent GP injury surgery and were labeled with AR 5 days post-injury. The tibias were harvested 2 days post-AR labeling and 1 week post-injury, the unlabeled area below the GP represents a new bone forming area [26] (Figure 3K,L, dotted lines). The width of the new bone-forming area was much narrower in the injured GP than in the contralateral uninjured GP 1 and 3 weeks post-injury (Figure 3L and Figure S-3, respectively). The inhibition of bone formation was not limited to the injured site but was detected in the entire GP (Figure 3M). Furthermore, we found a decrease in the number of osteoclasts in the injured GP (Figure 3N-P and Figure S-4). GPs injured with a 30 G needle did not show the inhibition of bone formation 1 week post-injury (Figure S-2C-2E), indicating that the GP injury with a 30 G needle does not impair GP function significantly.

3.3 | Retinoid Signaling in Injured Gp

To investigate whether injured GP upregulates retinoid signaling, we examined the expression of *Aldh1a3*, a restricted

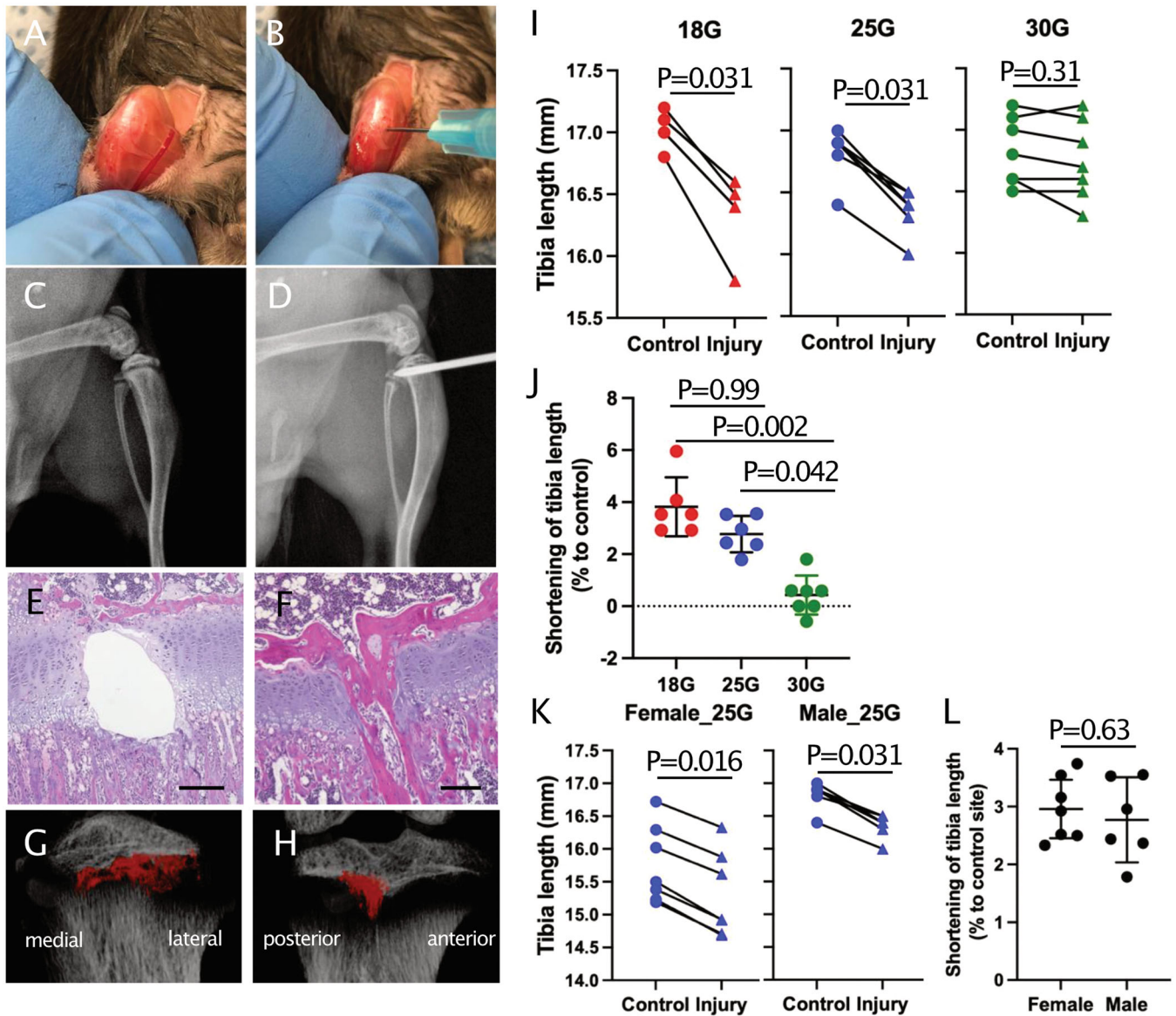


FIGURE 1 | Mouse GP injury models. The proximal GP in the right tibias of C57BL/6j male (A–L) and female (K and L) mice were subjected to GP injury surgery at 3.5–4 weeks of age. A–D, After the proximal tibia was exposed (A), the proximal tibia GP was pierced in a medial–lateral direction with a 30 G, 25 G, or 18 G needle (B). Radiological images before (C) and during (D) 25 G needle insertion. E–H, Histology of injured GP immediately (E) and 5 weeks (F) post-surgery. Bars are 200 μ m. The sagittal (G) and coronal (H) views of microCT images of the bone bridge (red regions) 5 weeks post-surgery with a 25 G needle. Images are representative of three subjects per group (E and F). I–L, the left (control, contralateral) and right (injury) tibias were harvested 5 weeks post-surgery, and the lengths of the tibias were measured (I and K). The percentage of the differences in bone length between the right and left tibias to the total length of the right bone was calculated (J and L). Graphs represent individual values ($n = 6–7$, $n = 6$ for 18 G and 25 G male, and 25 G female; $n = 7$ for 30 G male), average and standard deviation. The Wilcoxon matched-pairs test was used to compare control and injury groups in (I and K). The Kruskal–Wallis test was used to compare the groups with different sizes of needles in (J). The Mann–Whitney U test was used to compare female and male groups in (L).

enzyme for the synthesis of the most potent active form of retinoid (all-trans retinoic acid) [27] in the GP 1 and 3 days post-injury. We found that the injured GP contained more transcripts and proteins of *Aldh1a3* (Figure 4A–D, transcripts; Figure 4E,F, proteins). Furthermore, the expression of *Tgm2*, a direct target molecule of retinoid signaling [28], was also more evident in the GP 3 days post-injury compared to that in the uninjured GP (Figure 4G,H). Expression of *Aldh1a3* and *Tgm2* proteins was significantly higher in the hypertrophic zone in the injured GP than that in the control GP (Figure 4I and J).

3.4 | The Effects of the RAR γ Antagonist on GP Injuries

These results indicate that retinoid signaling is upregulated in the GP immediately post-injury; therefore, we investigated the significance of increased retinoid signaling activity in the GP injury. To do so, we administrated the selective antagonist of RAR γ , 7C described in WO 2005/066115 A2 from 1 day post-surgery (Figure 5A). Treatment with 7C decreased *Tgm2* expression (Figure S-5). The difference in growth

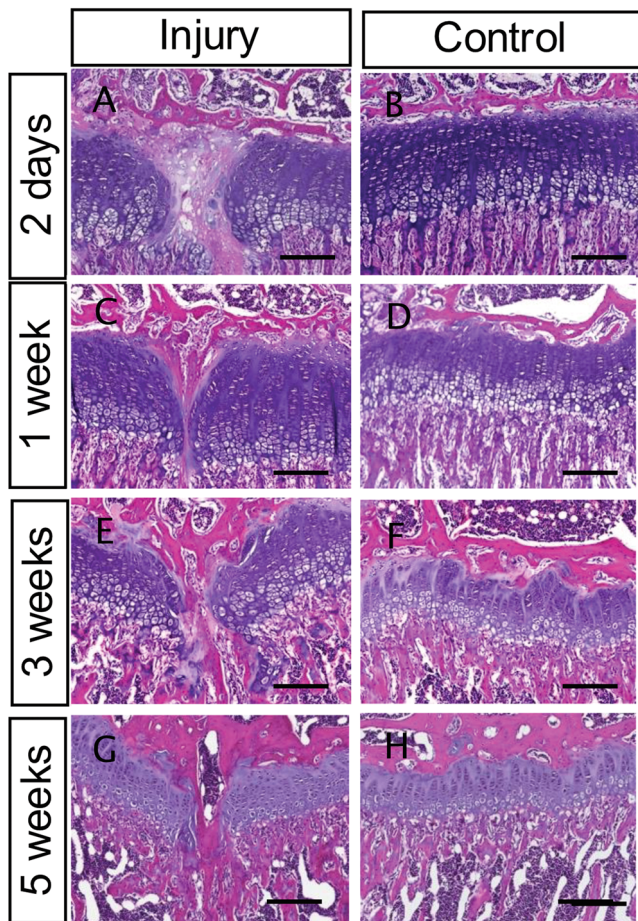


FIGURE 2 | Histological changes of the injured growth plate over time. The proximal GP in the right tibias of C57BL/6j male mice were subjected to GP injury surgery at 3.5 weeks of age. The right (A, C, E and G) and left (B, D, F and H) tibias were harvested 2 days (A and B), 1 week (C and D), 3 weeks (E and F) and 5 weeks (G and H) post-surgery ($n = 3$). The sagittal sections were stained with hematoxylin and eosin. Images are representative of three subjects per group. Bar are 200 μm .

between the injured and contralateral uninjured tibias was compared between the 7C and control vehicle groups. In female mice, treatment with 7C attenuated the differences in bone growth between the injured and contralateral bones both 10 and 20 days after surgery (Figure 5B,C). In male mice, the 7C effect was only significant 20 days post-surgery (Figure 5F). In all mice, the 7C treatment inhibited bone bridge formation (Figure 5D,G and Figure S-6). To examine the effects of treatment with 7C, bone-forming activity and osteoclast formation were measured in the female mice. A comparison of the untreated and 7C-treated injured GP showed that treatment with 7C increased the bone-forming activity (Figure 5H, injured cont vs. injured 7C) while it did not affect bone forming activity in uninjured GPs (Figure 5H, uninjured cont vs. uninjured 7C) and bone growth of the uninjured limb as previously reported [23, 29]. Treatment with 7C mitigated the injury-induced decrease in osteoclast formation (Figure 5I, injured cont vs injured 7C and Figure S-5), but did not affect cell proliferation as measured by EdU uptake (Figure 5J).

4 | Discussion

4.1 | GP Injury Mouse Models

In this study, we refined the GP injury mouse model and demonstrated that injuries in the GP induced with various needle sizes produced different healing results. The injury created with a 30 G needle was repaired without significant sequelae while the injury created with a 25 G needle resulted in bone bridge and LLD. The use of these two models would be helpful in studying differences in the mechanisms underlying GP repair and growth arrest, as the healing results were likely dependent on whether the needle diameter exceeded the width of the GP. Connecting the epiphyseal bone marrow to metaphyseal bone marrow may prevent integration of the edges of the damaged GP and lead to bone bridge formation. Therefore, the needle size should be adjusted for each mouse strain depending on the width of GP. Additionally, age at the time of surgery is important. At younger ages, when the secondary ossification center has not yet been established, complete repair is likely, even with a larger needle size. However, a needle smaller than the width of the GP may cause growth arrest at older ages because GP activity declines with age. The 25 G injury model has an injury line penetrating from epiphysis to metaphysis and consistently induces bone bridge and LLD. It is difficult to define this model according to the Salter-Harris classification: induction of consistent LLD is a feature of a Salter-Harris type IV, but the fracture line resembles a Salter-Harris type II. In contrast, the 30 G injury model confines the injury within the growth plate and may be considered a Salter-Harris type I fracture. Compared to the GP injury models previously reported [8, 12–14, 24], our GP injury model minimized damage to surrounding tissues such as the diaphysis of bone and articular cartilage, and thereby may not mimic the true clinical situation, which is a limitation of this model. However, due to the small size of mouse bones, it is difficult to replicate drill holes in the GP and surrounding tissues at exactly the same location and angle, which can be a biasing factor in experiments. Our models may be a valuable tool for the precise and detailed analysis of injury-induced functional and molecular changes in GP chondrocytes.

Sun et al. [30] have recently reported a sophisticated mouse GP injury model using compound tricolor reporter transgenic mice that can locate collagen 2-expressing and collagen 10 expressing areas. They used 2-week-old mice and made a defect, 0.5 mm in diameter and depth, from the metaphyseal bone to the hypertrophic zone [30]. Both our 25 G injury model and the tricolor model limit damage to the GP and consistently induce bone bridge formation and LLD induction. Our model minimizes damage to surrounding tissues and focuses on studying the direct response of growth plate chondrocytes to injury, and may be less clinically relevant than the tricolor model, which mimics a Salter-Harris type 2 fracture [30]. Relying on visual inspection of the growth plate without being guided by reporter indicators may result in less accurate positioning, but instead, any mice can be used in our model. The growth plate in 3–4 week old mice was clearly detected under the stereomicroscope (Figure S-1). We used 3–4 week old mice while a tricolor mouse model used 2 week old mice. One of the reasons we chose 3–4 weeks of age instead of 2 weeks of age is maturity of

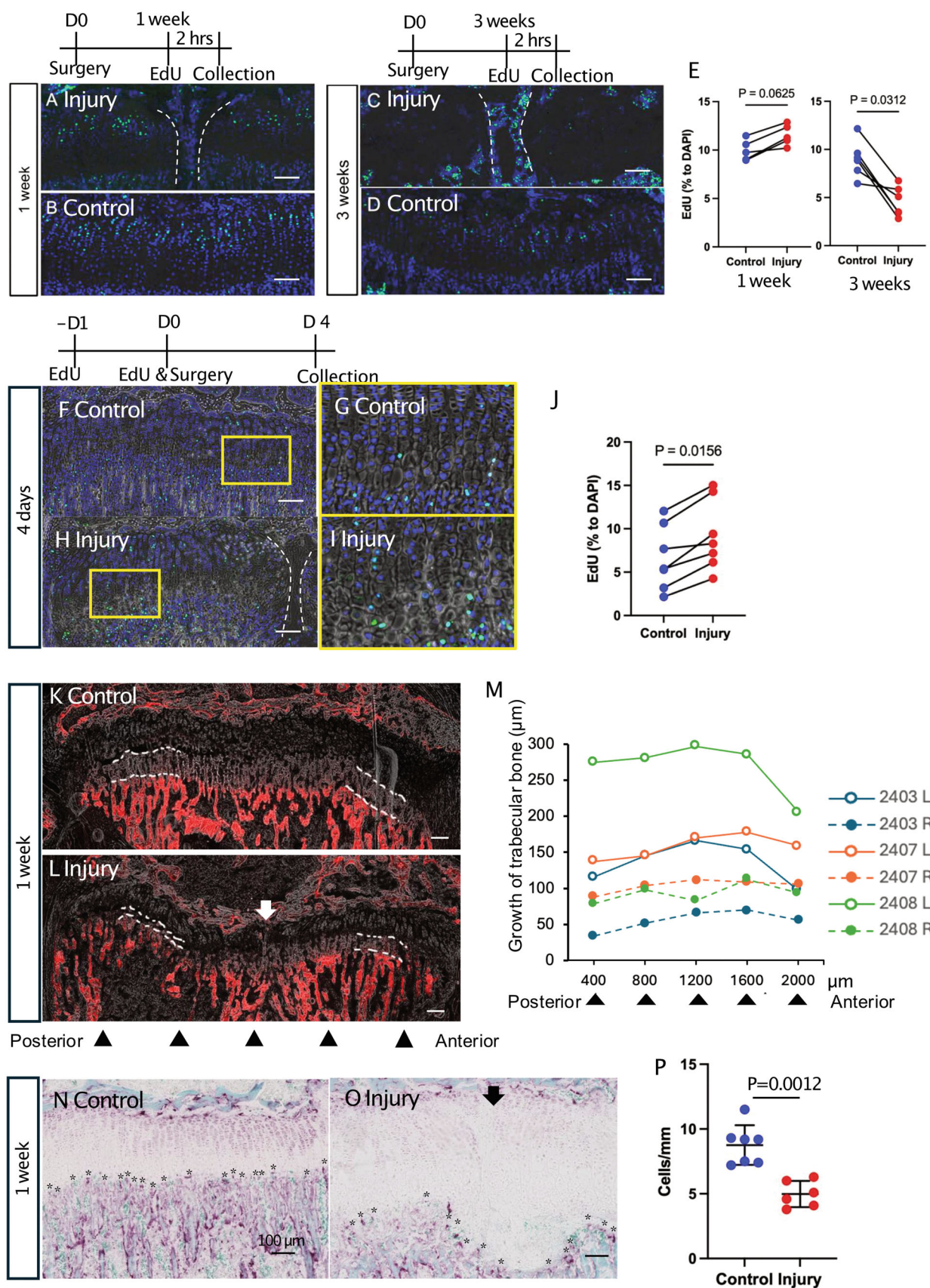


FIGURE 3 | Legend on next page.

the growth plate structure. The tibia in the 3-week-old mouse has well-demarcated GP structure as human tibia does so even in infants [31] while the tibia in the 2-week-old mouse does not establish a clear border between the secondary ossification center and the resting zone. Furthermore, the mode of renewal of GP chondrocytes by chondroprogenitors is still a neonatal type in the 2 week old GP [32], which may complicate the study of the cellular mechanisms of growth arrest. We also avoided complications that may arise from feeding transitions around weaning. Human has two growth phases (an early growth, 3-6 years; a spurt growth, 8-11/13 years) [31] while mice has an early growth until 2 weeks of age and then steady phase until 6 weeks of age [25]. To select the best mouse age in the GP injury model we should consider many factors such as growth rate, structure maturity, feeding status and injury type.

4.2 | Growth Arrest Caused by GP Injuries

This study demonstrated that the GP changes rapidly post-injury, preceding bone bridge formation. The changes found 1 week post-injury include slower cell turnover as well as decreased bone-forming activity and osteoclast number in the chondro-osseous junction in the metaphysis. These results indicated that the endochondral ossification process was impaired by the injury. In particular, a decrease in bone forming activity is directly associated with growth arrest [26]. The decrease in bone-forming activity continued 3 weeks post-injury (Figure S-1). Although bone formation was histologically evident 3 weeks post-injury, the connectivity of the bone bridge to the trabecular bones in the metaphysis was less evident than that 5 weeks post-injury (Figure 2E,G). These findings suggest that the inhibition of longitudinal bone growth occurs before the physical block of growth by the bone bridge. Since proliferative activity was not reduced 1 week post-injury, the reason for the decreased cell turnover is likely a delay in events required for bone replacement, including the terminal differentiation of chondrocytes, vascular invasion, absorption of calcified matrix and differentiation of osteoprogenitors. Another possibility is the inactivation of resting chondrocytes/chondroprogenitors that supply new chondrocytes to the GP, as previously suggested [15].

The initial treatment for GP injury currently monitoring injury healing, pain and growth imbalance [33]. If growth imbalance in a bone or between the affected and unaffected bones is predicted to cause severe LLD or skeletal deformity, temporal or permanent epiphysiodesis is a surgical option to slow or stop unaffected GP function [6, 9, 34]. Clinical and preclinical studies have explored intervention techniques and materials to stimulate GP repair and prevent LLD. Successful reduction in bone bridge formation by quick release of the antibody against vascular endothelial growth factor has been reported; however, the inhibition of growth disturbance by this treatment has not been identified [8]. The restoration of the GP injury-induced defect with the poly(ethylene glycol)-based resin with hydrogels containing glycosaminoglycans or TGF β 3 represents a positive action on the improvement of LLD [35]. The application of mesenchymal stem cells into injured GP has been studied clinically and preclinically; however, their effectiveness has not been concluded [36]. Our findings suggest the possibility that the amelioration of early impairment of the GP function may effectively prevent future growth arrest and imbalance.

4.3 | Upregulation of Retinoid Signaling in the GP Repair

We observed that retinoid signaling was acutely increased in injured GP. The activation of retinoid signaling in injured tissues is generally a pro-action for tissue repair, though not always [21]. Retinoid signaling supports the proliferation and differentiation of progenitor cells and/or reactivation of cell programs in various injured tissues [37]. The expression of ALDH1A3, a rate-limiting enzyme in the production of retinoic acid from vitamin A, increases in myofibroblasts surrounding acutely injured tubules in the kidney, leading to the stimulation of retinoid signaling that supports tubule repair for a short period [38, 39]. Upregulation of retinoid signaling has been also found immediately post-muscle injury [22, 40], and the administration of agonists for RAR γ stimulates muscle repair [22] and inhibits fat infiltration [40] in animal models. Furthermore, the clinical benefits of vitamin A derivatives in skin repair have been demonstrated [37].

FIGURE 3 | Characterization of early changes in injured GP. The proximal GP in the right tibiae of C57BL/6j female mice were subjected to GP injury surgery at 4 weeks of age. The left (control) and right (injury) tibiae were harvested 4 days (F–J), 1 week (A, B, E, and K–P) or 3 weeks (C–E) post-surgery. (A–E), Mice received the injection of EdU 2 h before collection of the tibiae 1 week (A, B, and E, $n = 5$) and 3 weeks (C–E, $n = 6$) post-surgery. The EdU-labeled cells were visualized (A–D), and the labeling index (EdU positive cells to DAPI cells) was calculated (E). The dotted line indicates the border of the injury area. Bars are 100 μ m. Graphs represent individual values ($n = 5$ for 1 week; $n = 6$ for 3 weeks), average, and standard deviation. The Wilcoxon matched-pairs test was used to compare control and injury groups (E). F–J, Mice received the EdU injection one day before surgery, and the left (control) and right (injury) tibiae were harvested 4 days after surgery. The EdU-labeled cells were visualized (F–I, $n = 7$), and the labeling index (EdU positive cells to DAPI cells) was calculated (J). The dotted line indicates the border of the injury area. Bars are 100 μ m. Graphs represent individual values ($n = 6$). The Wilcoxon matched-pairs test was used to compare control and injury groups. (K–M), Mice (IDs, 2403 (Female), 2407 (Male), and 2408 (Male)) received the injection of AR 2 days in before collection of the tibiae 1 week post-surgery. The alizarin-incorporated calcified materials were imaged under the fluorescence microscopy (K and L). A white arrow indicates the injury site. The unlabeled area below the hypertrophic zone between the dotted lines indicates newly formed trabecular bone. Bars are 200 μ m. The width of this region was measured with a gap of 400 μ m in the left (control, L) and right (injury, R) tibiae (M). N–O, Sagittal sections of the left (Control) and right (Injury) tibiae were subjected to tartrate-resistance acid phosphatase (TRAP) staining (N and O). A black arrow represents the injury site. Stars indicate TRAP-positive osteoclasts. Bars are 100 μ m. The number of osteoclasts at the chondro-osseous junction was counted. Graphs represent the individual values ($n = 7$), average, and standard deviation (P). The Mann-Whitney U test was used to compare control and injury groups.

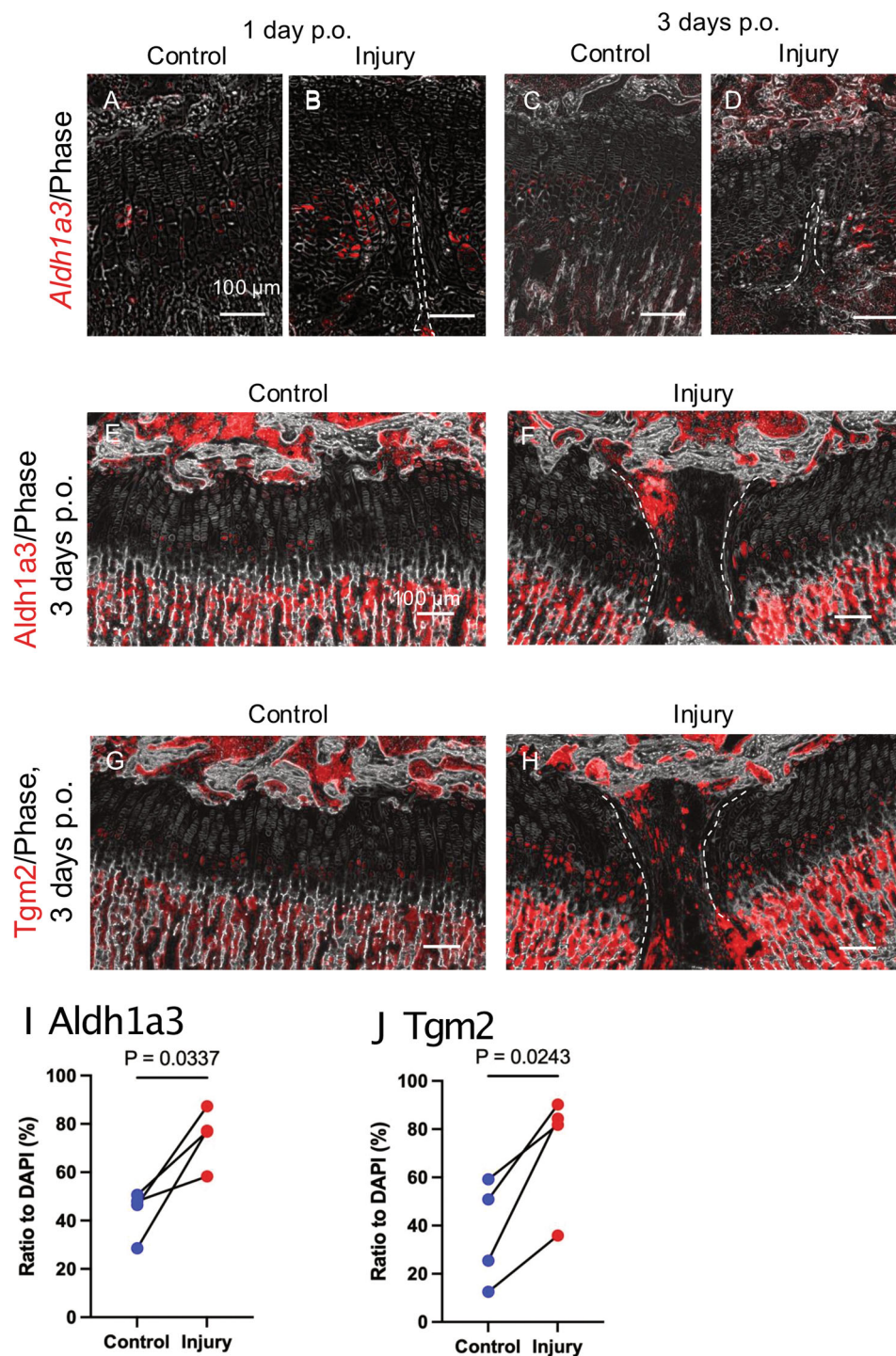


FIGURE 4 | Expression of retinoid signaling target molecules in injured growth plate. The proximal growth plate in the right tibiae of C57BL/6j female mice were subjected to GP injury surgery at 4 weeks of age. The left (control) and right (injury) tibiae were harvested 1 day (A and B) and 3 days (C–H) post-surgery. Sagittal sections were subjected to in situ hybridization for *Aldh1a3* transcripts (A–D, $n = 3$ per time point) or immunofluorescence staining for Aldh1a3 (E and F, $n = 4$) and Tgm2 (G and H, $n = 4$). Images were captured under the fluorescence microscope (red) and superimposed with phase contrast (gray) images. Images are representative of three (*Aldh1a3* transcripts) or four (Aldh1a3 and Tgm2 immunostaining) subjects. The ratio of number of Tgm2/Aldh1a3 positive cells to number of DAPI positive cells in the hypertrophic zone was calculated (I and J). Bars are 100 μm .

In contrast to the reparative effects of retinoid signaling in the aforementioned tissues, acute stimulation of retinoid signaling may be detrimental in the GP because administration of a RAR γ antagonist attenuated LLD induced by GP injury. Excess activation of retinoid signaling has negative effects on the skeleton,

including the deterioration of cortical bone thickness and quality [41] and the precocious closure of the GP [42]. Furthermore, retinoids inhibit mineralization of osteoblasts in culture [43], and vitamin A precursors suppress osteoclastogenesis [44]. These findings are consistent with the observations

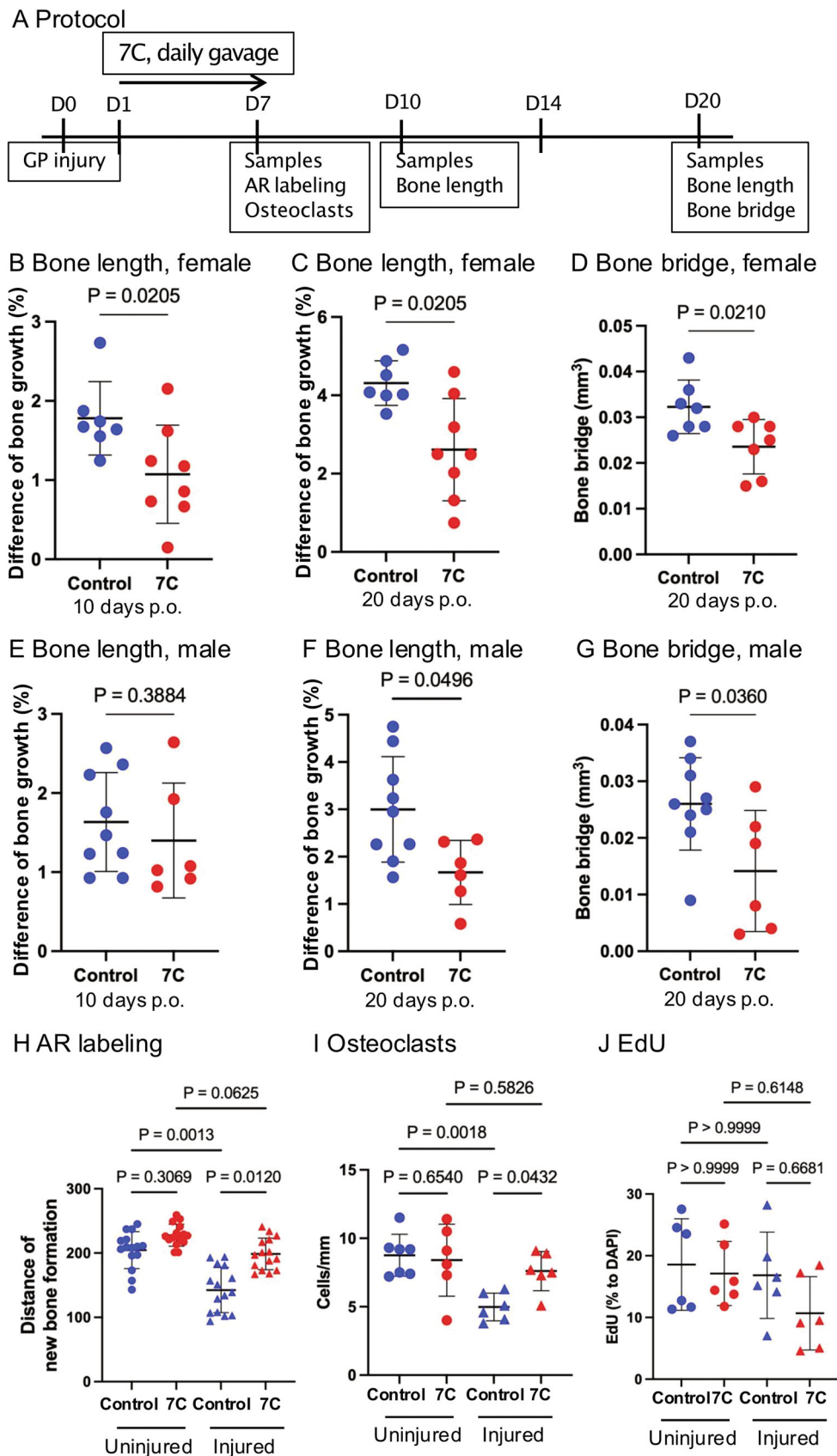


FIGURE 5 | Legend on next page.

that the injured GP having hyper activation of retinoid signaling showed a decrease in new bone formation and osteoclast formation and that the RAR γ antagonist counteracted such effects.

In contrast, treatment with retinoic acid strongly enhances the expression of hypertrophic traits and matrix mineralization in cultured avian chondrocytes [45]. Beta-carotene, a precursor of vitamin A and low concentration of retinoic acid stimulate the early differentiation of osteoblasts in culture [46]. The inactivation of RAR signaling in osteoblasts induces craniofacial deformities [47] and osteopetrotic conditions [48] in mice. Furthermore, retinoic acid has been reported to enhance RANKL-induced osteoclast formation [49], and RAR α -mediated signaling in bone marrow stromal cells supports osteoclast formation in culture [47]. These findings highlight the physiological importance of retinoid signaling in skeletal growth and maintenance. It is likely that retinoid signaling activated in the injured GP should be considered a pathological dysregulation but not an enhancement of its endogenous role, as seen in other tissues [37].

Our results suggest that 7 C, a RAR γ antagonist, may be useful in suppressing the hyperactivation of retinoid signaling in injured GPs, attenuating GP arrest and bridge formation, and correcting bone growth pathology. The 7 C was administered systemically for the first 7 days post-injury but did not completely block the injury-induced pathology. The 7 C regimen was set for 7 days, as there was no information on its toxicity beyond this period. Future studies should be conducted to optimize the treatment frequency, duration, and dosage to maximize therapeutic efficacy without causing significant side effects. Furthermore, targeted applications of 7 C should be investigated, for example, by experiments that demonstrate the complementary effect of 7 C loaded into polylactide nanoparticles with BMP2 on endochondral ossification in a mouse spinal fusion model [50]. The therapeutic action of the RAR γ antagonist (CD2665) has been also demonstrated by preventing the arrest of mouse GP induced by the antagonist of the hedgehog pathway, LDE225 (sonidegib), which has been developed for the treatment of variety of cancers including basal cell carcinomas and medulloblastomas [23]. On the other hand, the RAR γ agonist (palovarotene) has been approved as a clinical drug to inhibit heterotopic ossification in fibrodysplasia ossificans progressiva, a fatal genetic skeletal disease [51]. Our findings and those of others indicate that pharmacological control of retinoid signaling may be an effective treatment for a

variety of skeletal disorders involving the endochondral ossification process.

Author Contributions

M.M. and M.I. designed the experiments. M.M., N.T., T.H., K.U., M.E.I. and M.I. conducted experiments; M.M., S.O., N.I., J.M.A., M.E.I. and M.I. evaluated the results; and M.E.I. and M.I. prepared the manuscript. All authors have read and approved the final submitted manuscript.

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Ethics Statement

All animal experiments were approved by the Institutional Animal Care and Use Committees at Children's Hospital of Philadelphia (959) and University of Maryland, Baltimore (0120005 and 00000024).

Translational study.

Conflicts of Interest

MEI is a spouse of MI.

Data Availability Statement

All data will be made available from the corresponding author M.I. upon reasonable request.

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FIGURE 5 | The actions of 7 C, the retinoid antagonist on stimulation of GP repair. A The proximal growth plate in the right tibiae of C57BL/6j female (B–D and H–J) and male (E–G) mice were subjected to GP injury surgery at 4 weeks of age. 7 C was orally administered daily from one day to 7 days after surgery. B–G The lengths of the right and left tibiae of control (no treatment) and 7 C groups (female and male mice) were measured at the time of surgery, and 10 days (B and E, $n = 7$ female and $n = 9$ male for control; $n = 8$ female and $n = 6$ male for 7 C) and 20 days (C and F, $n = 9$ male for control and $n = 6$ male for 7 C) post-surgery. The increased length of left and right tibiae from the time of surgery was calculated and the difference in the increased length between left and right tibiae was evaluated as the percentage to the left tibia length (B, C, E and F). The bone bridge volume was analyzed 20 days post-surgery (D and G). Graphs represent individual values, average and standard variation. The Mann-Whitney U test was used to compare control and 7 C groups. H–J Mice received AR injections 5 days post-surgery and further received EdU injections 2 hours before euthanasia 7 days post-surgery ($n = 6$). Age-matched mice without GP injury also received AR and EdU in the same manner ($n = 7$ for vehicle control; $n = 6$ for 7 C). The left and right tibiae were harvested 7 days post-surgery. Alizarin staining, TRAP staining and EdU staining images of the sagittal sections of the samples were captured. The alizarin-unlabeled distance below the hypertrophic zone (H, 2–3 sections per individual sample), the number of osteoclasts at the chondro-osseous junction (I) and the EdU labeling index (J) were measured. The Kruskal-Wallis test was used to compare four groups.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supplementary Information Matsuoka R1. Supplement Figures R2.